System Architecture Specification

This document outlines the core decoupled design of modern hybrid retrieval-augmented generation systems.

The key invariants include zero credential persistence, ephemeral in-memory context lifecycles, and strict network egress isolation.

Vector Storage and Retrieval Constraints

Vector databases must enforce server-side scope filters on every query to guarantee multi-tenant and session-level isolation.

Reciprocal rank fusion fuses sparse lexical representations with dense semantic embeddings.